

Día 3

1. BUILD SUCCESS de Maven.

```
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops\app> .\mvnw.cmd clean verify
[INFO] Scanning for projects...
[INFO]
[INFO] -----< org.springframework.samples:spring-petclinic >-----
[INFO] --- clean:3.5.0:clean (default-clean) @ spring-petclinic ---
[INFO]
[INFO] --- clean:3.5.0:clean (default-clean) @ spring-petclinic ---
[INFO]
[INFO] --- enforcer:3.6.2:enforce (enforce-java) @ spring-petclinic ---
[INFO] Rule 0: org.apache.maven.enforcer.rules.version.RequireJavaVersion passed
[INFO]
[INFO] --- spring-javaformat:0.0.47:validate (default) @ spring-petclinic ---
[INFO]
[INFO] --- checkstyle:3.6.0:check (nohttp-checkstyle-validation) @ spring-petclinic ---
[INFO] You have 0 Checkstyle violations.
[INFO]
[INFO] --- jacoco:0.8.14:prepare-agent (default) @ spring-petclinic ---
[INFO] argLine set to "-javaagent:C:\Users\luis2\m2\repository\org\jacoco\org.jacoco.agent\0.8.14\org.jacoco.agent-0
runtime.jar-destfile=C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops\app\ta
jacoco.exec"
[INFO]
[INFO] --- git-commit-id:9.0.2:revision (default) @ spring-petclinic ---
[INFO] dotGitDirectory 'C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops\.git'
[INFO] collected git.build.user.name with value luisfloresc
[INFO] collected git.build.user.email with value luis2001dc@gmail.com
[INFO] collected git.branch with value develop
[INFO] collected git.commit.id.describe with value c9e8a6e-dirty
[INFO] collected git.commit.id.describe-short with value c9e8a6e-dirty
[INFO] collected git.commit.id with value c9e8a6ee7a9227b092c1912b6b3891692c569b2d
[INFO] collected git.commit.id.abbrev with value c9e8a6e
[INFO] collected git.dirty with value true
[INFO] collected git.commit.user.name with value Hugo Alexander Pepinosa Moreno
[INFO] collected git.commit.user.email with value alexispepinosa20@gmail.com
[INFO] collected git.commit.message.full with value dev
[INFO] collected git.commit.message.short with value dev
[INFO] collected git.commit.time with value 2026-06-03T00:21:06-05:00
[INFO] collected git.commit.author.time with value 2026-06-03T00:21:06-05:00
[INFO] collected git.commit.committer.time with value 2026-06-03T00:21:06-05:00
[INFO] Scanning for projects...
Proyecto-Devops\app)
2026-06-03T09:47:37.926-05:00 INFO 22184 --- [main] o.s.s.petclinic.vet.VetControllerTests : No active profile s
lling back to 1 default profile: "default"
2026-06-03T09:47:38.448-05:00 INFO 22184 --- [main] o.s.b.t.m.w.SpringBootMockServletContext : Initializing Spring
ispatcherServlet ''
2026-06-03T09:47:38.449-05:00 INFO 22184 --- [main] o.s.t.web.servlet.TestDispatcherServlet : Initializing Serve
2026-06-03T09:47:38.450-05:00 INFO 22184 --- [main] o.s.t.web.servlet.TestDispatcherServlet : completed initializ
2026-06-03T09:47:38.456-05:00 INFO 22184 --- [main] o.s.s.petclinic.vet.VetControllerTests : Started VetControll
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.868 s -- in org.springframework.samples.petclinic.vet
ntrollerTests
[INFO] Running org.springframework.samples.petclinic.vet.VetTests
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.032 s -- in org.springframework.samples.petclinic.vet
sts
[INFO]
[INFO] Results:
[INFO]
[WARNING] Tests run: 59, Failures: 0, Errors: 0, Skipped: 2
[INFO]
[INFO]
[INFO] --- jacoco:0.8.14:report (report) @ spring-petclinic ---
[INFO] Loading execution data file C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops\ap
get\jacoco.exec
[INFO] Analyzed bundle 'petclinic' with 22 classes
[INFO]
[INFO] --- jar:3.4.2:jar (default-jar) @ spring-petclinic ---
[INFO] Building jar: C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops\app\target\spr
clinic-4.0.0-SNAPSHOT.jar
[INFO]
[INFO] --- spring-boot:4.0.3:repackage (repackage) @ spring-petclinic ---
[INFO] Replacing main artifact C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops\app\t
spring-petclinic-4.0.0-SNAPSHOT.jar with repackaged archive, adding nested dependencies in BOOT-INF/.
[INFO] The original artifact has been renamed to C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Pro
Devops\app\target\spring-petclinic-4.0.0-SNAPSHOT.jar.original
[INFO]
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 46.353 s
[INFO] Finished at: 2026-06-03T09:47:41-05:00
[INFO] -----
```

2. docker build -t spring-petclinic-devops:local ./app exitoso.

```
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> docker build -t spring-petclinic
s:local ./app
[+] Building 207.4s (18/18) FINISHED docker:desktop-
=> [internal] load build definition from Dockerfile
=> => transferring dockerfile: 634B
=> [internal] load metadata for docker.io/library/eclipse-temurin:17-jre-alpine
=> [internal] load metadata for docker.io/library/maven:3.9.6-eclipse-temurin-17
=> [auth] library/maven:pull token for registry-1.docker.io
=> [auth] library/eclipse-temurin:pull token for registry-1.docker.io
=> [internal] load .dockerignore
=> => transferring context: 2B
=> [builder 1/6] FROM docker.io/library/maven:3.9.6-eclipse-temurin-17@sha256:29a1658b1f3078e07c2b17f7b519b45eb47f65d9628e8
=> => resolve docker.io/library/maven:3.9.6-eclipse-temurin-17@sha256:29a1658b1f3078e07c2b17f7b519b45eb47f65d9628e87eac45a8
=> => sha256:fc06d68d71ba375a9ceed165c0c04b09b1a8cc193146888663b005660fc25013 156B / 156B
=> => sha256:4712df479f9c080dd6f61d3feab07459351189e9267f950f68e9b252e6ef558c 358B / 358B
=> => sha256:58c3491a14ebc973960cbda9f2eea22537785a3cb07e81487bd1115bad4a8278 852B / 852B
=> => sha256:45f480637770dbb740623c0c7f827d44bbac9204faf3d07b4135d54a2bee043 9.48MB / 9.48MB
=> => sha256:67f99c2688af1d0eced6d61dad057d7beffe543a8e8370d5cd3fe8682e059a3 19.00MB / 19.00MB
=> => sha256:a3ba11f7aaead962216812fe84aee9061aeabac6932f0274a1d204eb96e8e8 734B / 734B
=> => sha256:c97285723537ab7921fca4d081c256e501adfaa8992d04637992075f4cea392 173B / 173B
=> => sha256:d59fd278c1b4ffe1727be6ccba42125e8f4db57e660e795cce19889d2c776457 145.10MB / 145.10MB
=> => sha256:5e5d1bcc5440d3a24f4a620704b9e687b4163c6c872fcc8e812e200c9bbac58 17.46MB / 17.46MB
=> => sha256:4a023cab5400feb5c1ab725beb8345dd0b0e3200314004b56677a5ee2e8c86cf 30.44MB / 30.44MB
=> => extracting sha256:4a023cab5400feb5c1ab725beb8345dd0b0e3200314004b56677a5ee2e8c86cf
=> => extracting sha256:5e5d1bcc5440d3a24f4a620704b9e687b4163c6c872fcc8e812e200c9bbac58
=> => extracting sha256:d59fd278c1b4ffe1727be6ccba42125e8f4db57e660e795cce19889d2c776457
=> => extracting sha256:c97285723537ab7921fca4d081c256e501adfaa8992d04637992075f4cea392
=> => extracting sha256:a3ba11f7aaead962216812fe84aee9061aeabac6932f0274a1d204eb96e8e8
=> => extracting sha256:67f99c2688af1d0eced6d61dad057d7beffe543a8e8370d5cd3fe8682e059a3
=> => extracting sha256:45f480637770dbb740623c0c7f827d44bbac9204faf3d07b4135d54a2bee043
=> => extracting sha256:4712df479f9c080dd6f61d3feab07459351189e9267f950f68e9b252e6ef558c
=> => extracting sha256:58c3491a14ebc973960cbda9f2eea22537785a3cb07e81487bd1115bad4a8278
=> [stage-1 1/4] FROM docker.io/library/eclipse-temurin:17-jre-alpine@sha256:b0ae54a36f82e04dc6c45e40ca5c55762e20b9a0858ee4
=> => resolve docker.io/library/eclipse-temurin:17-jre-alpine@sha256:b0ae54a36f82e04dc6c45e40ca5c55762e20b9a0858ee457faf557d
=> => sha256:62425ad408e60d4c02718b125eb89dcd25069329b4733c049e2bfcca5c362569 2.28kB / 2.28kB
=> => sha256:982cdd985fb0f69f446d1579a7183d36288be91e53d912ddff0b23e63cb35020 128B / 128B
=> => sha256:1404f74f9fc080dd6f61d3feab07459351189e9267f950f68e9b252e6ef558c 47.23MB / 47.23MB
=> => sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb 3.86MB / 3.86MB
=> => sha256:c9352f4130d700690e156fa229f4f10961b67a764941770f6064b0b7f1ce4130 16.86MB / 16.86MB
=> => extracting sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb
=> => extracting sha256:1404f74f9fc080dd6f61d3feab07459351189e9267f950f68e9b252e6ef558c
=> => extracting sha256:982cdd985fb0f69f446d1579a7183d36288be91e53d912ddff0b23e63cb35020
=> => extracting sha256:62425ad408e60d4c02718b125eb89dcd25069329b4733c049e2bfcca5c362569
=> [internal] load build context
=> => transferring context: 1.30MB
=> [stage-1 2/4] RUN addgroup -S appgroup && adduser -S appuser -G appgroup
=> [stage-1 3/4] WORKDIR /app
=> [builder 2/6] WORKDIR /app
=> [builder 3/6] COPY pom.xml .
=> [builder 4/6] RUN mvn dependency:go-offline -B 1
=> [builder 5/6] COPY src ./src
=> [builder 6/6] RUN mvn package -DskipTests
=> [stage-1 4/4] COPY --from=builder /app/target/*.jar app.jar
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:9606a79658f2d3875ac983436cb7632a5c5db249158086d3183a4b4790f888368
=> => exporting config sha256:db2f124ea7111fd23bc095474bb0b675c5b4ea17dd606ecc456a4b83a1c21534
=> => exporting attestation manifest sha256:8f692f19dc4895b964f270823f42861773c939397f219518d8db0d476763686e
=> => exporting manifest list sha256:017b3cf2ac4a510276f93aa49669be159f4f5b5c6f926b1f7c8fb99c900d2bab
=> => naming to docker.io/library/spring-petclinic-devops:local
=> => unpacking to docker.io/library/spring-petclinic-devops:local
```

```
s:local ./app
[+] Building 207.4s (18/18) FINISHED docker:desktop-
=> [stage-1 1/4] FROM docker.io/library/eclipse-temurin:17-jre-alpine@sha256:b0ae54a36f82e04dc6c45e40ca5c55762e20b9a0858ee4
=> => resolve docker.io/library/eclipse-temurin:17-jre-alpine@sha256:b0ae54a36f82e04dc6c45e40ca5c55762e20b9a0858ee457faf557d
=> => sha256:62425ad408e60d4c02718b125eb89dcd25069329b4733c049e2bfcca5c362569 2.28kB / 2.28kB
=> => sha256:982cdd985fb0f69f446d1579a7183d36288be91e53d912ddff0b23e63cb35020 128B / 128B
=> => sha256:1404f74f9fc080dd6f61d3feab07459351189e9267f950f68e9b252e6ef558c 47.23MB / 47.23MB
=> => sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb 3.86MB / 3.86MB
=> => sha256:c9352f4130d700690e156fa229f4f10961b67a764941770f6064b0b7f1ce4130 16.86MB / 16.86MB
=> => extracting sha256:6a0ac1617861a677b045b7ff88545213ec31c0ff08763195a70a4a5adda577bb
=> => extracting sha256:1404f74f9fc080dd6f61d3feab07459351189e9267f950f68e9b252e6ef558c
=> => extracting sha256:982cdd985fb0f69f446d1579a7183d36288be91e53d912ddff0b23e63cb35020
=> => extracting sha256:62425ad408e60d4c02718b125eb89dcd25069329b4733c049e2bfcca5c362569
=> [internal] load build context
=> => transferring context: 1.30MB
=> [stage-1 2/4] RUN addgroup -S appgroup && adduser -S appuser -G appgroup
=> [stage-1 3/4] WORKDIR /app
=> [builder 2/6] WORKDIR /app
=> [builder 3/6] COPY pom.xml .
=> [builder 4/6] RUN mvn dependency:go-offline -B 1
=> [builder 5/6] COPY src ./src
=> [builder 6/6] RUN mvn package -DskipTests
=> [stage-1 4/4] COPY --from=builder /app/target/*.jar app.jar
=> exporting to image
=> => exporting layers
=> => exporting manifest sha256:9606a79658f2d3875ac983436cb7632a5c5db249158086d3183a4b4790f888368
=> => exporting config sha256:db2f124ea7111fd23bc095474bb0b675c5b4ea17dd606ecc456a4b83a1c21534
=> => exporting attestation manifest sha256:8f692f19dc4895b964f270823f42861773c939397f219518d8db0d476763686e
=> => exporting manifest list sha256:017b3cf2ac4a510276f93aa49669be159f4f5b5c6f926b1f7c8fb99c900d2bab
=> => naming to docker.io/library/spring-petclinic-devops:local
=> => unpacking to docker.io/library/spring-petclinic-devops:local
```

3. Docker ps

PS C:\Users\luis2> docker ps						
CONTAINER ID	IMAGE	NAMES	COMMAND	CREATED	STATUS	PORTS
78b1e251ea6a	spring-petclinic-devops:local		"java -jar app.jar"	5 minutes ago	Up 5 minutes (healthy)	0.0.0.0
:8080->8080/tcp, [::]:8080->8080/tcp		eloquent_nobel				

4. curl http://localhost:8080/actuator/health con StatusCode: 200.

```
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> curl http://localhost:8080/actuator/health

StatusCode      : 200
StatusDescription : 
Content         : {123, 34, 103, 114...}
RawContent      : HTTP/1.1 200
                  Transfer-Encoding: chunked
                  Keep-Alive: timeout=60
                  Connection: keep-alive
                  Content-Type: application/vnd.spring-boot.actuator.v3+json
                  Date: Wed, 03 Jun 2026 14:56:10 GMT

                  {"groups":...
Headers         : {[Transfer-Encoding, chunked], [Keep-Alive, timeout=60], [connection, keep-alive],
                  [Content-Type, application/vnd.spring-boot.actuator.v3+json]...}
RawContentLength : 49

PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> Invoke-RestMethod http://localhost:8080/actuator/health

groups      status
-----
{liveness, readiness} UP
```

DIA 4

```
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> minikube start --cpus=4 --memory=3000
minikube v1.38.1 on Microsoft Windows 10 Home 22H2
* Automatically selected the docker driver
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
* Using Docker Desktop driver with root privileges
* Starting "minikube" primary control-plane node in "minikube" cluster
* Pulling base image v0.0.50 ...
* Downloading Kubernetes v1.35.1 preload ...
  > preloaded-images-k8s-v18-v1...: 272.45 MiB / 272.45 MiB 100.00% 14.23 M
  > gcr.io/k8s-minikube/kicbase...: 519.58 MiB / 519.58 MiB 100.00% 9.17 Mi
* Creating docker container (CPUs=4, Memory=3000MB) ...
* Preparing Kubernetes v1.35.1 on Docker 29.2.1 ...
* Configuring bridge CNI (Container Networking Interface) ...
* Verifying Kubernetes components...
  * Using image gcr.io/k8s-minikube/storage-provisioner:v5
  * Enabled addons: storage-provisioner, default-storageclass
  * Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops>
```

```
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> minikube addons enable metrics-server
metrics-server is an addon maintained by Kubernetes. For any concerns contact minikube on GitHub.
You can view the list of minikube maintainers at: https://github.com/kubernetes/minikube/blob/master/OWNERS
  * Using image registry.k8s.io/metrics-server/metrics-server:v0.8.1
  * The 'metrics-server' addon is enabled
```

```
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> kubectl get pods -n kube-system
NAME                                READY   STATUS    RESTARTS   AGE
coredns-7d764666f9-m42dg           1/1     Running   0           2m5s
etcd-minikube                       1/1     Running   0           2m11s
kube-apiserver-minikube             1/1     Running   0           2m11s
kube-controller-manager-minikube    1/1     Running   0           2m11s
kube-proxy-j2dgg                   1/1     Running   0           2m5s
kube-scheduler-minikube             1/1     Running   0           2m13s
metrics-server-9d74bb658-9nmgj      0/1     Running   0           27s
storage-provisioner                 1/1     Running   1 (103s ago) 2m9s
```

```
NAME                                READY   STATUS    RESTARTS   AGE
coredns-7d764666f9-m42dg           1/1     Running   0           2m5s
etcd-minikube                       1/1     Running   0           2m11s
kube-controller-manager-minikube    1/1     Running   0           2m11s
kube-proxy-j2dgg                   1/1     Running   0           2m5s
metrics-server-9d74bb658-9nmgj      0/1     Running   0           27s
storage-provisioner                 1/1     Running   1 (103s ago) 2m9s
namespace/staging created
namespace/prod created
configmap/spring-petclinic-config created
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> kubectl apply -f k8s\secret.yaml -n staging
deployment.apps/spring-petclinic created
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> kubectl apply -f k8s\service.yaml -n staging
service/spring-petclinic created
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> kubectl apply -f k8s\pdb.yaml -n staging
poddisruptionbudget.policy/spring-petclinic-pdb created
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> kubectl get pods -n staging
NAME                                READY   STATUS    RESTARTS   AGE
spring-petclinic-cc8d67975-2fj22    0/1     ContainerCreating   0           15s
spring-petclinic-cc8d67975-pd84t     0/1     ContainerCreating   0           15s
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> kubectl get deployment -n staging
NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
spring-petclinic                    0/2     2             0           15s
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> kubectl get svc -n staging
NAME                                TYPE           CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
spring-petclinic                    ClusterIP      10.100.47.102 <none>        80/TCP      16s
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> kubectl get hpa -n staging
NAME                                REFERENCE          TARGETS          MINPODS   MAXPODS   REPLICAS   AGE
spring-petclinic                    Deployment/spring-petclinic  cpu: <unknown>/60%  2          10          2           15s
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> kubectl get pdb -n staging
NAME                                MIN   AVAILABLE   MAX UNAVAILABLE   ALLOWED DISRUPTIONS   AGE
spring-petclinic-pdb                1     N/A          0                  0                      16s
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> kubectl rollout status deployment/spring-petclinic -n staging
Waiting for deployment "spring-petclinic" rollout to finish: 0 of 2 updated replicas are available...
Waiting for deployment "spring-petclinic" rollout to finish: 1 of 2 updated replicas are available...
deployment "spring-petclinic" successfully rolled out
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> kubectl port-forward svc/spring-petclinic 8082:80 -n staging
Forwarding from 127.0.0.1:8082 -> 8080
Forwarding from [::1]:8082 -> 8080
```

```
PS C:\Users\luis2> curl.exe http://localhost:8082/actuator/health
{"groups":["liveness","readiness"],"status":"UP"}
```

```
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> kubectl get pods -n staging
NAME                                READY   STATUS    RESTARTS   AGE
spring-petclinic-cc8d67975-2fj22    1/1     Running   0           10m
spring-petclinic-cc8d67975-pd84t     1/1     Running   0           10m
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> kubectl rollout status deployment/spring-petclinic -n staging
deployment "spring-petclinic" successfully rolled out
PS C:\Users\luis2\Desktop\Octavo Semestre\Fabrica de Software\Trabajo_7_dias\Proyecto-Devops> kubectl get hpa -n staging
NAME                                REFERENCE          TARGETS          MINPODS   MAXPODS   REPLICAS   AGE
spring-petclinic                    Deployment/spring-petclinic  cpu: 1%/60%      2          10          2           10m
```

DIA 5